

Axel Feldmann

(917)-691-5910 • axelf@csail.mit.edu • feldmann.nyc

GOAL	Looking for computer architecture and/or performance engineering roles in industry. I'm specifically interested in roles that involve working across the hardware/software interface.		
EDUCATION	Massachusetts Institute of Technology , Cambridge, Massachusetts		
	Ph.D. in Electrical Engineering and Computer Science		<i>anticipated</i> in December 2024
	• Advisor: Daniel Sanchez		
	S.M. in Electrical Engineering and Computer Science		May 2021
	• GPA: 5.0 / 5.0		
	Carnegie Mellon University , Pittsburgh, Pennsylvania		
	BS in Computer Science		May 2019
	• GPA: 3.9 / 4.0		
	Selected coursework: computer architecture, operating systems, compilers, algorithms, computer networks, numerical algorithms		
SKILLS	Computer architecture research, hardware prototyping/development, parallel programming models, analytical and simulation-based performance modeling, hardware-software codesign. C++, C, Python, CUDA, x86/64 assembly, Julia, Rust, PyTorch, Triton, Pandas, cuBLAS, cuSPARSE, Verilog		
HONORS & AWARDS	Mathworks Fellowship		2024
	IEEE Micro Top Picks		2022
EXPERIENCE			
	Research Assistant		September 2019 – present
	MIT CSAIL - Cambridge, MA		
	• Currently working on designing hardware accelerators for sparse linear algebra, specifically focused on scientific computing. • Designed <i>Spatula</i> and <i>Azul</i>, hardware accelerators for sparse linear algebra capable of outperforming GPU baselines by gmean 47× and gmean 242× respectively. • Worked on designing hardware accelerators for Fully Homomorphic Encryption (FHE), a computationally expensive cryptographic system that enables arithmetic operations on encrypted data. • Designed <i>F1</i>, the first proposed ASIC accelerator capable of executing entire FHE programs. • Developed compilation techniques that enabled <i>F1</i> and <i>CraterLake</i> (our second-generation accelerator design) to achieve 5000× speedups over CPU baselines.		
	Intern		Summer 2022
	Cerebras Systems - Sunnyvale, CA		
	• Worked on the CSL (Cerebras Systems Language) team to designing new abstractions for programming the Cerebras Wafer Scale Engine. • Designed and built a complete MPI-style collectives library for CSL programmers.		
	Undergraduate Research Assistant		2018 – 2019
	Computer Organization Research Group - Pittsburgh, PA		
	• Worked with Prof. Nathan Beckmann on <i>Livia</i>, a system architecture for data centric computing. • Created zsim-based simulation infrastructure to evaluate our proposed architecture. • Wrote applications to effectively utilize <i>Livia</i>'s novel hardware features.		
	Systems Software Intern		Summer 2018
	NVIDIA - Santa Clara, CA		
	• Improved display driver performance for existing and upcoming Tegra SoCs. • Reduced kernel test time by 30% via improved thread synchronization.		
	Software Engineering Intern		Summer 2017
	Yahoo, Flurry Analytics - Sunnyvale, CA		
	• Created webapp to help users design metrics API queries. • Re-engineering User Acquisition Analysis (UAA) features on the Flurry data platform.		

COMPUTER ARCHITECTURE PAPERS	<u>Azul: An Accelerator for Sparse Iterative Solvers Leveraging Distributed On-Chip Memory</u> Axel Feldmann, Courtney Golden, Yifan Yang, Joel S. Emer, Daniel Sanchez, <i>to appear at MICRO 2024</i>. <u>Spatula: A Hardware Accelerator for Sparse Matrix Factorization</u> Axel Feldmann, Daniel Sanchez, <i>MICRO 2023</i>. <u>An Architecture to Accelerate Computation on Encrypted Data</u> Axel Feldmann*, Nikola Samardzic*, Aleksander Krastev, Srinivas Devadas, Ronald Dreslinski, Chris Peikert, Daniel Sanchez, <i>IEEE Micro Top Picks 2022</i>. <u>CraterLake: A Hardware Accelerator for Efficient Unbounded Computation on Encrypted Data</u> Nikola Samardzic, Axel Feldmann, Aleksander Krastev, Nathan Manohar, Nicholas Genise, Srinivas Devadas, Chris Peikert, Daniel Sanchez, <i>ISCA 2022</i>. <u>F1: A Fast and Programmable Accelerator for Fully Homomorphic Encryption</u> Axel Feldmann*, Nikola Samardzic*, Aleksander Krastev, Srinivas Devadas, Ron Dreslinski, Chris Peikert, Daniel Sanchez, <i>MICRO 2021</i>. <u>Livia: Data-Centric Computing Throughout the Memory Hierarchy</u> Eliot Lockerman, Axel Feldmann, Mohammad Bakhshalipour, Alexandru Stanescu, Shashwat Gupta, Daniel Sanchez, and Nathan Beckmann, <i>ASPLOS 2020</i>.	
GPU PROGRAMMING PAPERS	<u>FeatUp: A model-agnostic framework for features at any resolution</u> Stephanie Fu, Mark Hamilton, Laura Brandt, Axel Feldmann, Zhoutong Zhang, and William T. Freeman, <i>ICLR 2024</i>. <u>DsDm: Model-Aware Dataset Selection with Datamodels</u> Logan Engstrom, Axel Feldmann, and Aleksander Madry, <i>ICML 2024</i>. <u>MiniFold: Simple, Fast and Accurate Protein Structure Prediction</u> Jeremy Wohlwend, Mateo Reveiz, Matt McPartlon, Axel Feldmann, Wengong Jin, Regina Barzilay, <i>under submission to Neurips 2024</i>.	
TEACHING ASSISTANT	MIT 6.5900: Computer Systems Architecture CMU 15-410: Operating System Design and Implementation CMU 15-213: Introduction to Computer Systems	Fall 2024 (ongoing) 2018-2019 2017-2018

[Resume compiled on 2024-08-13]